

SIMATS ENGINEERING
CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

Assessment Tool Weightage: 40%

Duration: 90 minutes

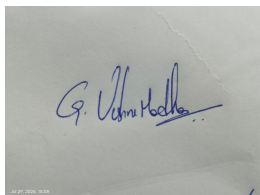
Marks: 25

Code of Conduct:

I G Vishnu Madhav (192425192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in blue ink on a white piece of paper. The signature appears to be 'G. Vishnu Madhav'.

Scenario-Based Questions

CO-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate	Shows correct relationships with minor gaps	Relationships are weak, unclear, or	No meaningful relationships established

		relationships between concepts		partially incorrect	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Name: G. Vishnu Madhav
Regno: 192425192
Course Code: MLA0408
Course Name: Deep Learning

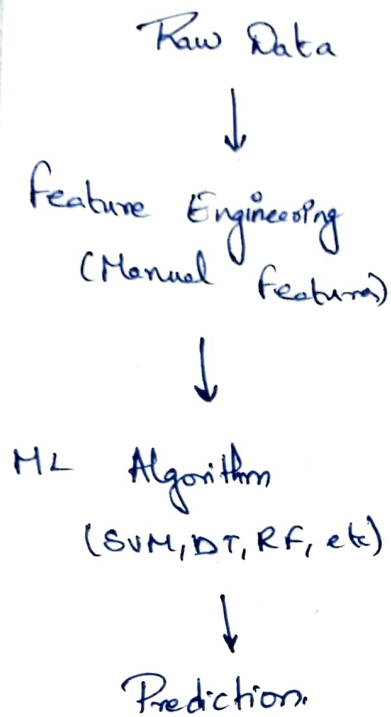
CO2 AT3

1. Differentiate b/w Machine Learning and Deep Learning.

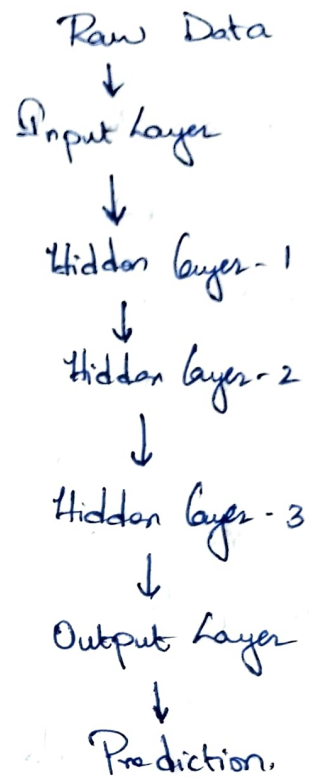
Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and make predictions without being explicitly programmed. Deep Learning (DL) is a subset of Machine Learning that uses artificial neural networks with multiple hidden layers to automatically learn complex patterns from data.

Feature	Machine Learning	Deep Learning
1. Architecture	Use algorithms like Decision Tree, SVM, KNN and Random forest.	Uses Artificial Neural Network with multiple hidden layers.
2. Data Requirement	Works well with small and medium-sized dataset.	Requires large amount of data for better accuracy.
3. Feature extraction	Features are selected manually by experts.	Features are learned automatically from raw data.
4. Training Time	Faster training and testing.	Requires more training & powerful GPUs.
5. Applications	Mainly used for structured data.	Best for image, speech & text data.

Machine Learning Architecture



Deep Learning Architecture



Real World examples

ML:

- * Spam Email Detection
- * Credit Card Fraud Detection

DL:

- * Face Recognition
- * Self-driving Cars.

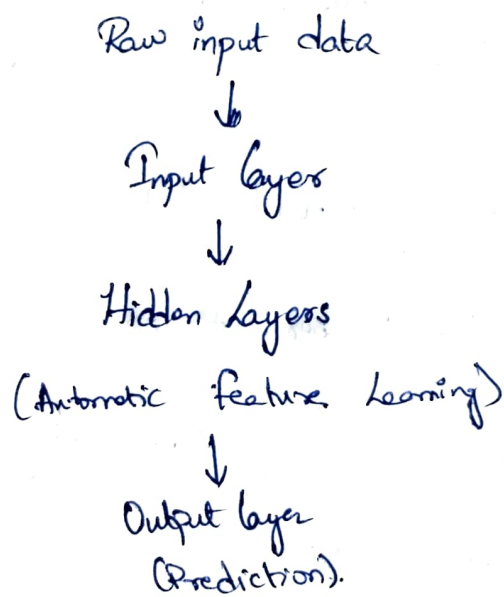
Conclusion:

ML is Suitable for structured data with few samples
Whereas Deep Learning is Suitable for large and complex
datasets because it automatically learns features and provide
higher accuracy.

2. Explain the Concept of Representation Layer Learning.

Representation Learning is a technique in Deep Learning where a neural network automatically learns useful features from raw data without manual feature engineering. The hidden layers extract features step by step starting from simple patterns and gradually learning complex representations. This improves the model's accuracy and reduces human effort.

Architecture



The hidden layers automatically learn important features from the input data, making this model suitable for solving complex problems.

Influence of Width & Depth.

Width refers to the numbers of neurons in each hidden layer, while Depth refers to the no. of hidden layers in a neural Network.

Feature	Width	Depth.
Def	No. of Neurons	No. of hidden layers
Learning Capacity	Learns more features in 1 layer	Learns complex hierarchical features
Computational Complexity	Moderate increase in Computation	Higher Computational and training time
Model performance	Improves accuracy for simple & medium tasks	Improves accuracy for complex task like image & speech recognition.

* Representation learning extracts features automatically.

* Width increases the number of neurons in a layer

* Depth increases the no. of hidden layers.

* More width improves feature learning capacity.

* More depth helps learn complex patterns.

Conclusion:

Representation learning enables automatic feature extraction

A balanced combination of network width & depth improves Capacity.

3. Compare ReLU, Leaky ReLU (LReLU) & ELU (Exponential Linear Unit) activation functions.

Activation functions are mathematical functions used in neural networks to introduce non-linearity. They help the network learn complex patterns by deciding whether a neuron should be activated. ReLU, Leaky ReLU and ELU:

1. ReLU

$$* f(x) = 0, \text{ if } x < 0$$

$$* f(x) = x, \text{ if } x \geq 0$$

Advantages

- * Simple & fast
- * Reduces the vanishing gradient problem.
- * Widely used in DL models.

Disadvantages

- * Can suffer from dying ReLU problem, when neurons stop learning.

Suitable Use Case

- * Image classification
- * Convolutional Neural Networks (CNN).

2. Leaky ReLU [LReLU]

$$* f(x) = 0.01x, \text{ if } x < 0$$

$$* f(x) = x, \text{ if } x \geq 0.$$

Advantages

- * Prevents the dying ReLU problem.
- * Allows a small gradient for -ve inputs.
- * Improves learning in DL.

Disadvantages

- * Slightly more computationally expensive than ReLU.

Suitable Use Case

- * Deep neural Network
- * Image Recognition.

3. ELU [EXponential LINEAR UNIT]

- * $f(x) = \alpha (e^x - 1)$, if $x < 0$.
- * $f(x) = x$, if $x \geq 0$.

Advantages:

- * Fast Convergence during training.
- * Produces Smoother learning.
- * Improves Model accuracy.

Disadvantages:

- * Requires more computational due to e function.

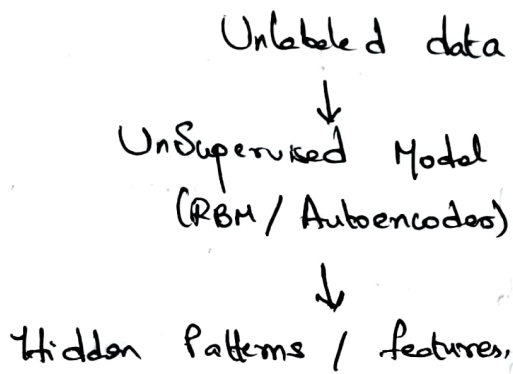
Suitable Use Case

- * Medical image analysis
- * Speech Recognition
- * NLP.

4. Concept of Unsupervised Learning in Neural Networks.

Unsupervised Learning is a type of ML in which the model learns patterns and relationships from unlabeled data. Unlike supervised learning, it does not require target outputs. It mainly uses feature extraction, clustering, dimensionality reduction, and anomaly detection.

Architecture



Restricted Boltzmann Machine (RBM)

RBM is an unsupervised neural network with 1 visible layer & 1 hidden layer. It learns hidden features by identifying relationships in the input data.

Working Principle

- * Input data is given to the visible layer
- * Hidden layer extracts important features
- * The network updates its weights to reduce reconstruction errors.

Auto Encoders

This is a neural network that learns a compressed representation of data and reconstructs the original output. It consists of Encoder, Latent Space & Decoder.

Working Principle

- * Encoder compresses the input layers
- * Latent Space stores important features
- * Decoder reconstructs the input with minimum loss.

Feature	RBM	Auto Encoders
Objective	learns hidden features from data	Learns compressed representation & reconstruction.
Architecture	Visible Layer + Hidden Layer	Encoder + Latent Space + Decoder
Training	Contrastive Divergence	Backpropagation
Application	Recommendation Systems, Feature extraction	Image denoising, data compression, anomaly detection.

— Thank You —